Department of Electrical and Computer Engineering
Queen's University

ELEC-271 Digital Systems

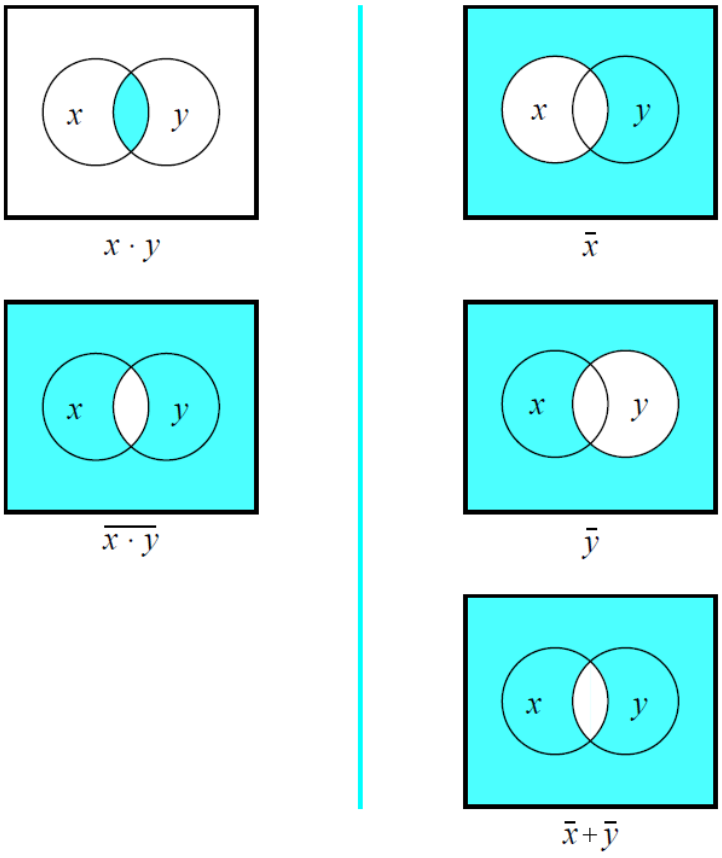
Solutions to Problem Set 1

1. Use algebraic manipulation to prove that $x + yz = (x + y)(x + z)$. This is the distributive rule, identity 12b in Section 2.5 of textbook.

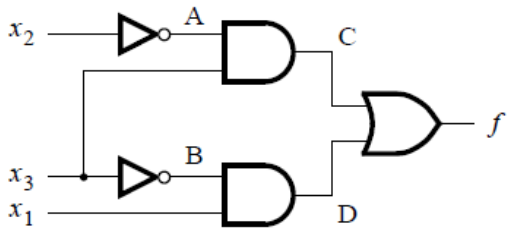
The proof is as follows:

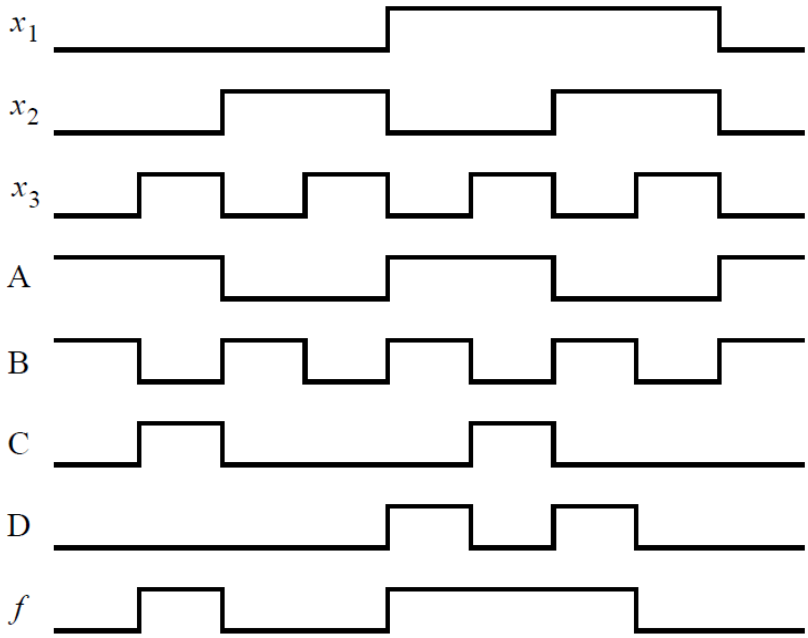
$$
\begin{aligned}
(x + y).(x + z) &= xx + xz + yx + yz && 12a \\
&= x + xz + yx + yz && 7a \\
&= x\,(1 + z + y) + yz && 12a \\
&= x.1 + yz && 5b \\
&= x + yz && 6a
\end{aligned}
$$

2. Use the Venn diagram to prove DeMorgan's theorem, identity 15a in Section 2.5 of textbook.



3. Draw a timing diagram for the following circuit. Show the waveforms that can be observed on all wires in the circuit.





4. Use algebraic manipulation to show that for three input variables x_1, x_2, and x_3:

$$\sum m(1,2,3,4,5,6,7) = x_1 + x_2 + x_3$$

Starting with the canonical sum-of-products for f:

$$
\begin{aligned}
f &= \overline{x}_1\overline{x}_2x_3 + \overline{x}_1x_2\overline{x}_3 + \overline{x}_1x_2x_3 + x_1\overline{x}_2\overline{x}_3 + x_1\overline{x}_2x_3 + x_1x_2\overline{x}_3 + x_1x_2x_3 \\
&= x_1(\overline{x}_2\overline{x}_3 + \overline{x}_2x_3 + x_2\overline{x}_3 + x_2x_3) + x_2(\overline{x}_1\overline{x}_3 + \overline{x}_1x_3 + x_1\overline{x}_3 + x_1x_3) \\
&\quad + x_3(\overline{x}_1\overline{x}_2 + \overline{x}_1x_2 + x_1\overline{x}_2 + x_1x_2) \\
&= x_1(\overline{x}_2(\overline{x}_3 + x_3) + x_2(\overline{x}_3 + x_3)) + x_2(\overline{x}_1(\overline{x}_3 + x_3) + x_1(\overline{x}_3 + x_3)) \\
&\quad + x_3(\overline{x}_1(\overline{x}_2 + x_2) + x_1(\overline{x}_2 + x_2)) \\
&= x_1(\overline{x}_2 \cdot 1 + x_2 \cdot 1) + x_2(\overline{x}_1 \cdot 1 + x_1 \cdot 1) + x_3(\overline{x}_1 \cdot 1 + x_1 \cdot 1) \\
&= x_1(\overline{x}_2 + x_2) + x_2(\overline{x}_1 + x_1) + x_3(\overline{x}_1 + x_1) \\
&= x_1 \cdot 1 + x_2 \cdot 1 + x_3 \cdot 1 \\
&= x_1 + x_2 + x_3
\end{aligned}
$$

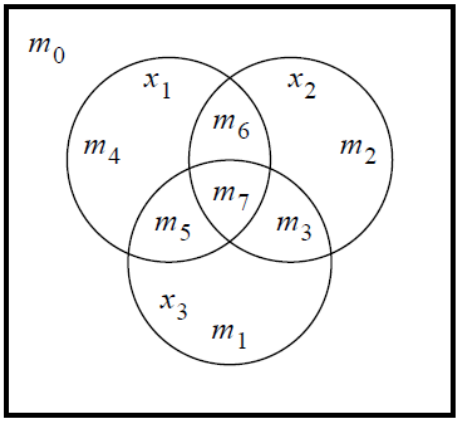
5. Use algebraic manipulation to find the minimum product-of-sums expression for the function:

$$f = (x_1 + x_2 + x_3)(x_1 + \overline{x}_2 + x_3)(\overline{x}_1 + \overline{x}_2 + x_3)(x_1 + x_2 + \overline{x}_3)$$

Derivation of the minimum product-of-sums expression:

$$
\begin{aligned}
f &= (x_1 + x_2 + x_3)(x_1 + \overline{x}_2 + x_3)(\overline{x}_1 + \overline{x}_2 + x_3)(x_1 + x_2 + \overline{x}_3) \\
&= ((x_1 + x_2) + x_3)((x_1 + x_2) + \overline{x}_3)(x_1 + (\overline{x}_2 + x_3))(\overline{x}_1 + (\overline{x}_2 + x_3)) \\
&= (x_1 + x_2)(\overline{x}_2 + x_3)
\end{aligned}
$$

6. Show the location of all minterms in a three-variable Venn diagram.



7. Design the simplest product-of-sums expression for the function:

$$f(x_1, x_2, x_3) = \prod M(0,1,5,7)$$

The simplest POS implementation of the function is:

$$\begin{aligned} f &= (x_1 + x_2 + x_3)(x_1 + x_2 + \overline{x}_3)(\overline{x}_1 + x_2 + \overline{x}_3)(\overline{x}_1 + \overline{x}_2 + \overline{x}_3) \\ &= ((x_1 + x_2) + x_3)((x_1 + x_2) + \overline{x}_3)((\overline{x}_1 + x_3) + x_2)((\overline{x}_1 + x_3) + \overline{x}_2) \\ &= (x_1 + x_2)(\overline{x}_1 + \overline{x}_3) \end{aligned}$$

8. Derive the simplest sum-of-products expression for the function:

$$f(x_1, x_2, x_3, x_4) = x_1\overline{x}_3\overline{x}_4 + x_2\overline{x}_3x_4 + x_1\overline{x}_2\overline{x}_3$$

The simplest SOP expression for the function is:

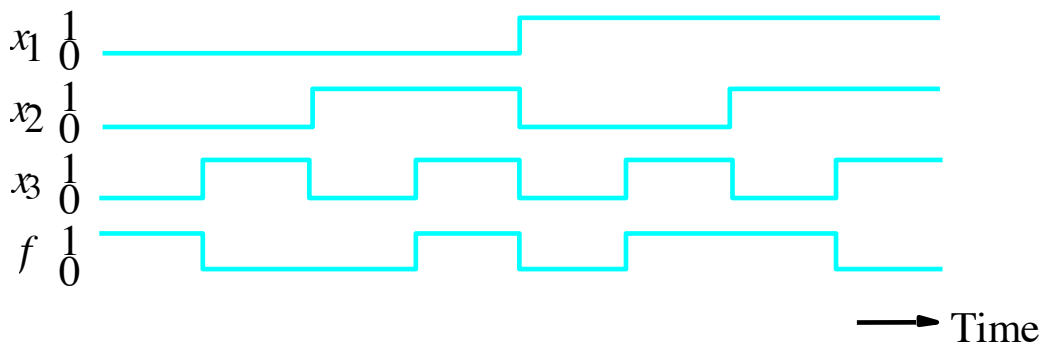
$$\begin{aligned} f(x_1, x_2, x_3, x_4) &= x_1\overline{x}_3\overline{x}_4 + x_2\overline{x}_3x_4 + x_1\overline{x}_2\overline{x}_3 \\ &= x_1\overline{x}_3(\overline{x}_2 + \overline{x}_4) + x_2\overline{x}_3x_4 = x_1\overline{x}_3(\overline{x_2x_4}) + x_2\overline{x}_3x_4 \\ &= \overline{x}_3(x_1\overline{x_2x_4} + x_2x_4) = \overline{x}_3(x_1 + x_2x_4) \\ &= x_1\overline{x}_3 + x_2\overline{x}_3x_4 \end{aligned}$$

9. Design the simplest circuit that has three inputs, x_1, x_2, and x_3, which produces an output value of 1 whenever exactly one or two of the input variables have the value 1; otherwise, the output has to be 0.

The function, f, of this circuit is equal to 0 when either none of the inputs or all three inputs are equal to 0. Otherwise, f is equal to 1. (this can be inferred from the truth table – draw the truth table). Therefore, using the POS form, the desired circuit can be realized as:

$$f(x_1, x_2, x_3) = \prod M(0,7) = (x_1 + x_2 + x_3)(\overline{x}_1 + \overline{x}_2 + \overline{x}_3)$$

10. For the following timing diagram, synthesize the function $f(x_1, x_2, x_3)$ in the simplest sum-of-products form.



The truth table that corresponds to the timing diagram:

x_1	x_2	x_3	f
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

The simplest SOP expression is:

$$f = \overline{x}_1\overline{x}_2\overline{x}_3 + \overline{x}_1 x_2 x_3 + x_1\overline{x}_2 x_3 + x_1 x_2\overline{x}_3$$

11. Design a circuit with output f and inputs x_1, x_0, y_1, and y_0. Let $X = x_1x_0$ be a number where the four possible values of X, namely, 00, 01, 10, and 11, represent the four numbers, 0, 1, 2, and 3, respectively. Similarly, $Y = y_1y_0$ represents another number with the same four possible values. The output f should be 1 if the numbers represented by X and Y are equal. Otherwise, f should be 0.

(a) Show the truth table for f.
(b) Synthesize the simplest possible product-of-sums expression for f.

(a)

x_1	x_0	y_1	y_0	f
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

(b)

$$f(x_1, x_0, y_1, y_0) = \prod M(1,2,3,4,6,7,8,9,11,12,13,14) = M_1 M_2 M_3 M_4 M_6 M_7 M_8 M_9 M_{11} M_{12} M_{13} M_{14}$$

$$M_2 M_3 = (x_1 + x_0 + \bar{y}_1 + y_0)(x_1 + x_0 + \bar{y}_1 + \bar{y}_0) = (x_1 + x_0 + \bar{y}_1)$$

$$M_6 M_7 = (x_1 + \bar{x}_0 + \bar{y}_1 + y_0)(x_1 + \bar{x}_0 + \bar{y}_1 + y_0) = (x_1 + \bar{x}_0 + \bar{y}_1)$$

$$M_8 M_9 = (\bar{x}_1 + x_0 + y_1 + y_0)(\bar{x}_1 + x_0 + y_1 + \bar{y}_0) = (\bar{x}_1 + x_0 + y_1)$$

$$M_{12} M_{13} = (\bar{x}_1 + \bar{x}_0 + y_1 + y_0)(\bar{x}_1 + \bar{x}_0 + y_1 + \bar{y}_0) = (\bar{x}_1 + \bar{x}_0 + y_1)$$

➔ $f(x_1, x_0, y_1, y_0) = M_1 M_2 M_3 M_4 M_6 M_7 M_8 M_9 M_{11} M_{12} M_{13} M_{14}$

$$= M_1(x_1 + x_0 + \bar{y}_1) M_4(x_1 + \bar{x}_0 + \bar{y}_1)(\bar{x}_1 + x_0 + y_1) M_{11}(\bar{x}_1 + \bar{x}_0 + y_1) M_{14}$$

$$= M_1(x_1 + \bar{y}_1) M_4(\bar{x}_1 + y_1) M_{11} M_{14}$$

We can use M_3, M_6, M_9, and M_{12} again (Property 7a) ➔

$$f(x_1, x_0, y_1, y_0) = M_1 M_3(x_1 + \bar{y}_1) M_4 M_6(\bar{x}_1 + y_1) M_9 M_{11} M_{12} M_{14}$$

$$M_1 M_3 = (x_1 + x_0 + y_1 + \bar{y}_0)(x_1 + x_0 + \bar{y}_1 + \bar{y}_0) = (x_1 + x_0 + \bar{y}_0)$$

$$M_4 M_6 = (x_1 + \bar{x}_0 + y_1 + y_0)(x_1 + \bar{x}_0 + \bar{y}_1 + y_0) = (x_1 + \bar{x}_0 + y_0)$$

$$M_9 M_{11} = (\bar{x}_1 + x_0 + y_1 + \bar{y}_0)(\bar{x}_1 + x_0 + \bar{y}_1 + \bar{y}_0) = (\bar{x}_1 + x_0 + \bar{y}_0)$$

$$M_{12} M_{14} = (\bar{x}_1 + \bar{x}_0 + y_1 + y_0)(\bar{x}_1 + \bar{x}_0 + \bar{y}_1 + y_0) = (\bar{x}_1 + \bar{x}_0 + y_0)$$

➔

$$f(x_1, x_0, y_1, y_0) = (x_1 + x_0 + \bar{y}_0)(x_1 + \bar{y}_1)(x_1 + \bar{x}_0 + y_0)(\bar{x}_1 + y_1)(\bar{x}_1 + x_0 + \bar{y}_0)(\bar{x}_1 + \bar{x}_0 + y_0)$$

➔ *The simplest POS possible expression is:*

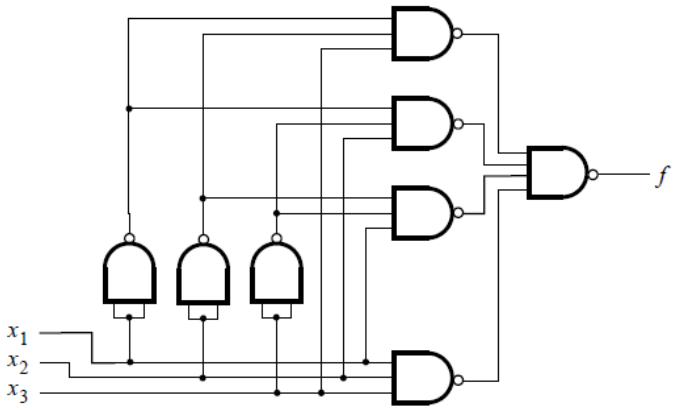
$$f(x_1, x_0, y_1, y_0) = (x_1 + \bar{y}_1)(\bar{x}_1 + y_1)(x_0 + \bar{y}_0)(\bar{x}_0 + y_0)$$

Please note that other approaches may exist.

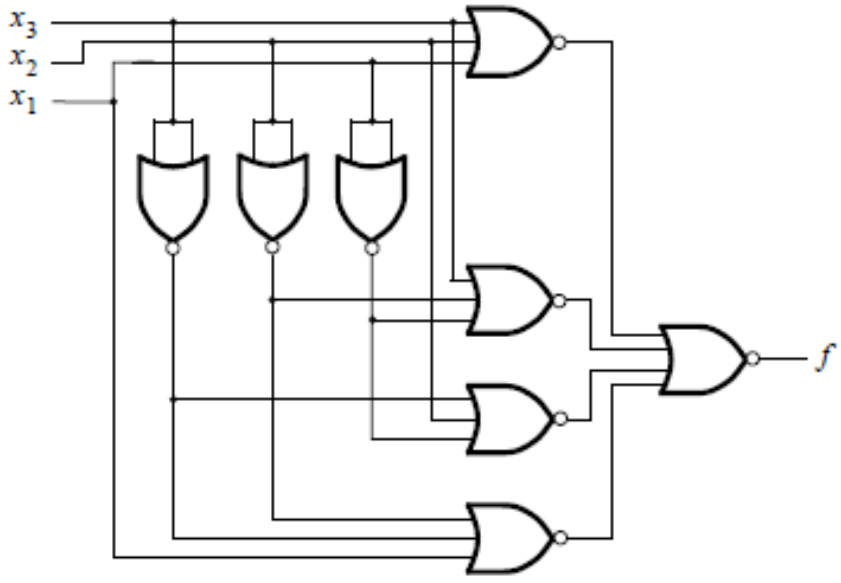
12. Implement the function for the following truth table using only NAND gates.

x_1	x_2	x_3	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

$$f = \bar{x}_1\bar{x}_2x_3 + \bar{x}_1x_2\bar{x}_3 + x_1\bar{x}_2\bar{x}_3 + x_1x_2x_3$$



(b) $$f = \prod M(0,3,5,6) = (x_1 + x_2 + x_3)(x_1 + \bar{x}_2 + \bar{x}_3)(\bar{x}_1 + x_2 + \bar{x}_3)(\bar{x}_1 + \bar{x}_2 + x_3)$$



13. Write the VHDL code to implement the function:

$$f(x_1, x_2, x_3) = \sum m(0,1,3,4,5,6).$$

The function can be specified by using the minterms as follows:

$$f = \bar{x}_1\bar{x}_2\bar{x}_3 + \bar{x}_1\bar{x}_2x_3 + \bar{x}_1x_2x_3 + x_1\bar{x}_2x_3 + x_1x_2x_3 + x_1x_2\bar{x}_3$$

The VHDL code is:

```
entity Q13 is
   port    (x1, x2, x3:     in bit;
             f:             out bit);
end entity Q13;

architecture logicfunc of Q13 is
begin

    f <= (NOT x1 AND NOT x2 AND NOT x3) OR (NOT x1 AND NOT x2 AND x3) OR
         (NOT x1 AND x2 AND x3) OR (x1 AND NOT x2 AND NOT x3) OR
         (x1 AND NOT x2 AND x3) OR (x1 AND x2 AND NOT x3) ;

end architecture logicfunc;
```

The simplest SOP expression for this function is:

$$f = \bar{x}_2 + x_1\bar{x}_3 + \bar{x}_1x_3$$

Using this expression, we can replace the statement that specifies f in the above VHDL code with the statement:

```
f <= NOT x2 OR (x1 AND NOT x3) OR (NOT x1 AND x3) ;
```

Another way of specifying the function is by using the maxterms, M_2 and M_7, in which case the VHDL statement would be:

```
f <= (x1 OR NOT x2 OR x3) AND (NOT x1 OR NOT x2 OR NOT x3) ;
```